How much stake is affected by a larger k

Comparison Epoch 228 with Epoch 616

1 Setup and notation

We analyze stake-pool snapshots from two specific epochs (228 and 616). Throughout this analysis, we hold the observed stake distribution $\{\sigma_i\}$ for each epoch fixed and vary only the protocol parameter k.

Goal. We compare the previous increase in k with the current proposal to increase k from 500 to 1000. For context, the previous change was announced in epoch 228 and took effect in epoch 234, raising k from 150 to 500 (a 3.33× increase). Our objective is to show how much stake was affected by that past change versus how much stake would be affected by the new proposal. To provide a fair comparison, we also calculate the affected stake under hypothetical scenarios where both increases have the same magnitude (either a 2× increase for both epochs, or a 3.33× increase for both epochs).

An important caveat to this comparison is that a large portion of the affected stake may belong to whales or centralized exchanges (CeXs). Consequently, this stake might not migrate from large pools to smaller ones. Instead, these large holders might open new pools to reallocate their stake or, in the worst-case scenario, partially or fully exit the staking ecosystem.

The figures reported here are intended solely to inform the community and do not imply any recommendation.

Formulas. The per-pool saturation *point* is

$$z_0(k) \;=\; \frac{T}{k}, \tag{1}$$

where T denotes the total supply. The saturation *level* is

$$s_i(k) \;=\; \frac{\sigma_i}{z_0(k)} \;=\; \frac{\sigma_i\, k}{T}, \tag{2}$$

where σ_i is the ADA staked at pool i.

Pool i is **oversaturated** if $s_i(k) > 1$, i.e. $\sigma_i > z_0(k)$. Stake above saturation (aggregated over pools in the file) is

$$E(k) \;=\; \sum_{i\,:\,\sigma_i>0} \max\{\sigma_i - z_0(k),\, 0\}. \tag{3}$$

We report $E(k)$ as a percentage of T and, separately, of S (total stakes).

2 Epoch 228

Table 1: Epoch 228 totals.

Quantity	Value
T (B ADA)	32.03
S (B ADA)	17.21
S/T	53.70 %
Pools with $\sigma_i > 0$	1,062.00

Table 2: Epoch 228 scenarios

Quantity	Previous ($k = 150$)	After ($k = 500$, $3.33\times$)	Hypothetical ($k = 300$, $2\times$)
$z_0(k)$ (M ADA)	213.53	64.06	106.77
Oversaturated pools (count)	0.00	109.00	69.00
Oversaturated pools (% of pools)	0.00 %	10.26 %	6.50 %
$E(k)$ - Stake above saturation (B ADA)	0.00	6.14	2.73
$E(k)$ (% of S)	0.00 %	35.71 %	15.88 %

3 Epoch 616

Table 3: Epoch 616 totals.

Quantity	Value
T (B ADA)	38.50
S (B ADA)	21.57
S/T	56.00 %
Pools with $\sigma_i > 0$	2,717.00

Table 4: Epoch 616 scenarios

Quantity	Previous ($k = 500$)	After ($k = 1000$, $2\times$)	Hypothetical ($k = 1667$, $3.33\times$)
$z_0(k)$ (M ADA)	77.00	38.50	23.10
Oversaturated pools (count)	8.00	212.00	344.00
Oversaturated pools (% of pools)	0.29 %	7.80 %	12.66 %
$E(k)$ - Stake above saturation (B ADA)	0.17	4.92	9.29
$E(k)$ (% of S)	0.77 %	22.79 %	43.07 %